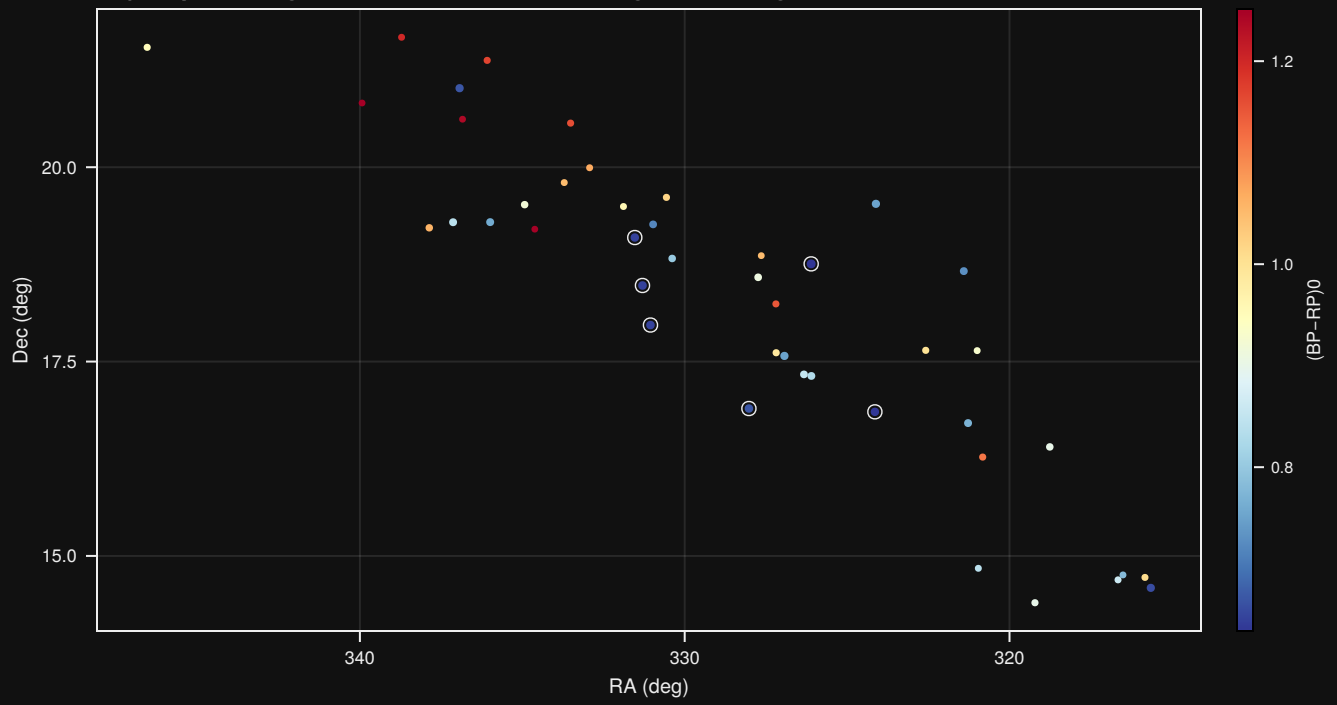
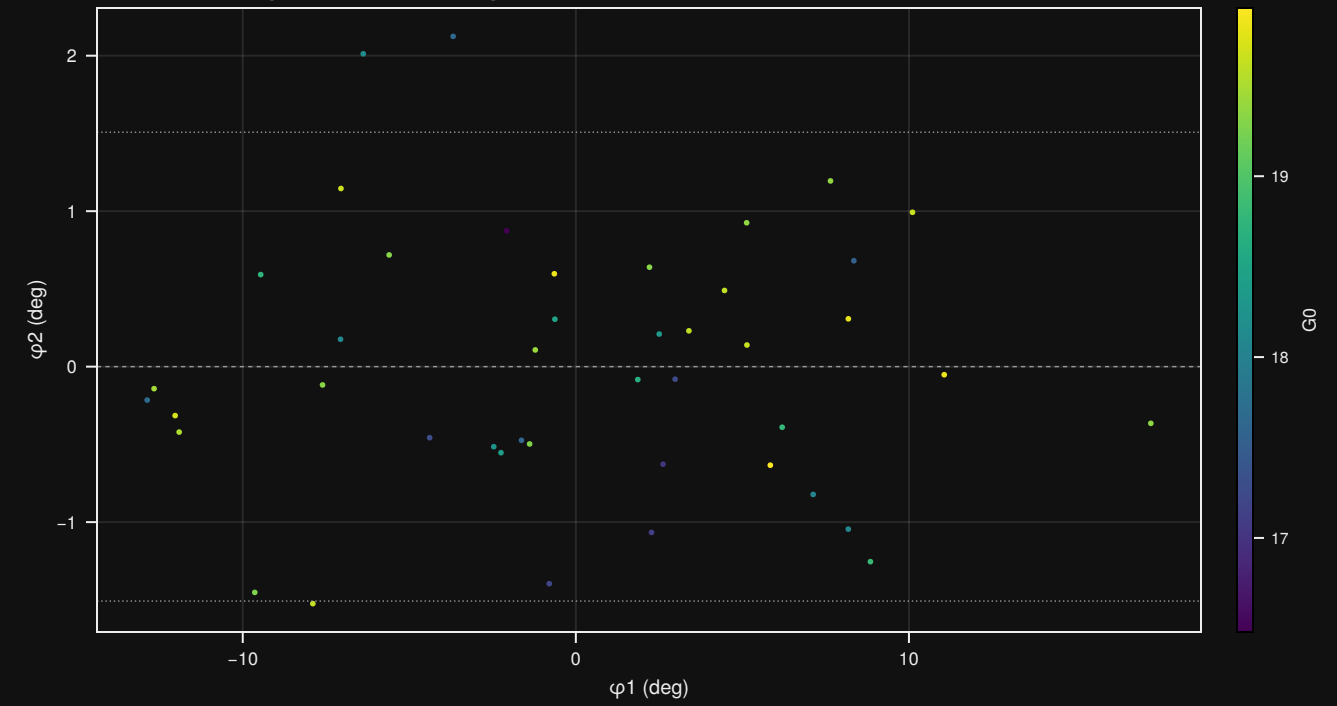
New-23

MEMBERSHIP	
Stars in catalog	44
with Gaia GSP-Phot	21 (48%)
with radial velocity	6
Stream id (sID)	79
GEOMETRY · great-circle fit	
Centroid RA, Dec	328.5, 18.7 deg
Centroid l, b	74.8, -27.4 deg
Pole RA, Dec	187.9, 66.6 deg
Length (φ1, 2–98%)	30.1 deg
Width σ(φ2), robust	0.75 deg
KINEMATICS	
pmRA* med ± σ	21.81 ± 5.36 mas/yr
pmDE med ± σ	-1.31 ± 1.62 mas/yr
μφ1* med ± σ	20.76 ± 4.94 mas/yr
μφ2 med ± σ	-6.60 ± 2.13 mas/yr
Parallax (cat.) med	0.266 ± 0.267 mas
Parallax (wtd, ZP-corr.)	0.317 ± 0.029 mas
Distance	d = 3.2 ± 0.3 kpc
Physical length	1.66 kpc
RV median ± σ(MAD)	-396.4 ± 19.6 km/s
RV gradient dV/dφ1	0.92 km/s/deg
PHOTOMETRY · POPULATIONS	
G0 range (median)	16.5–19.9 (19.0)
(BP−RP)0 median	0.90
GSP-Phot [M/H] (n=21)	-1.94 ± 0.42
GSP-Phot Teff median	5488 K
RUWE > 1.4	0.0 %
PM clusters (largest)	5 (16 stars)
IBATA+2024 · TABLE 3	
n catalog | paper	44 | 44
n_RV catalog | paper	6 | 6
<[Fe/H]> paper	-2.25
Stellar mass (paper)	120 Msun
Pole offset vs paper	0.2 deg

Sky map · catalog members (n = 44) · size ∝ brightness · ring = has RV

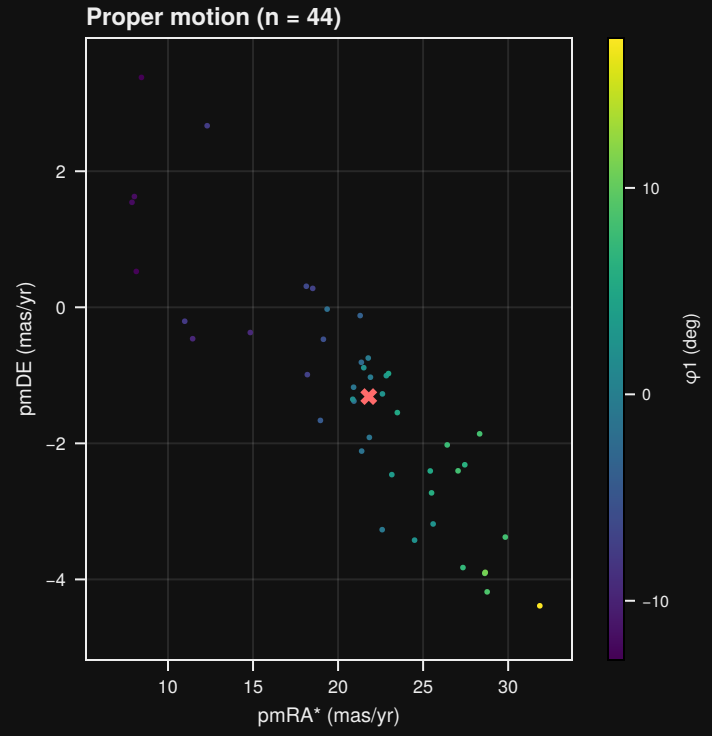


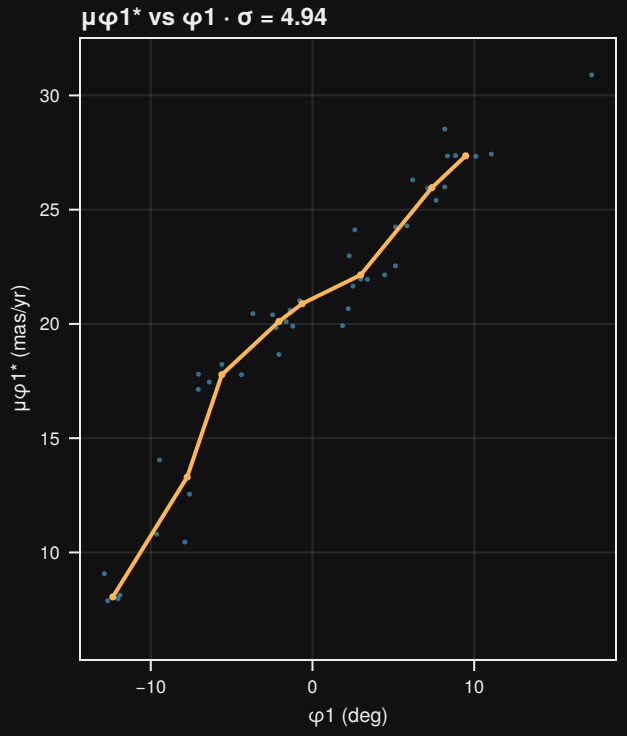
Stream frame (great-circle fit) · length 30.1° · width σ 0.75°

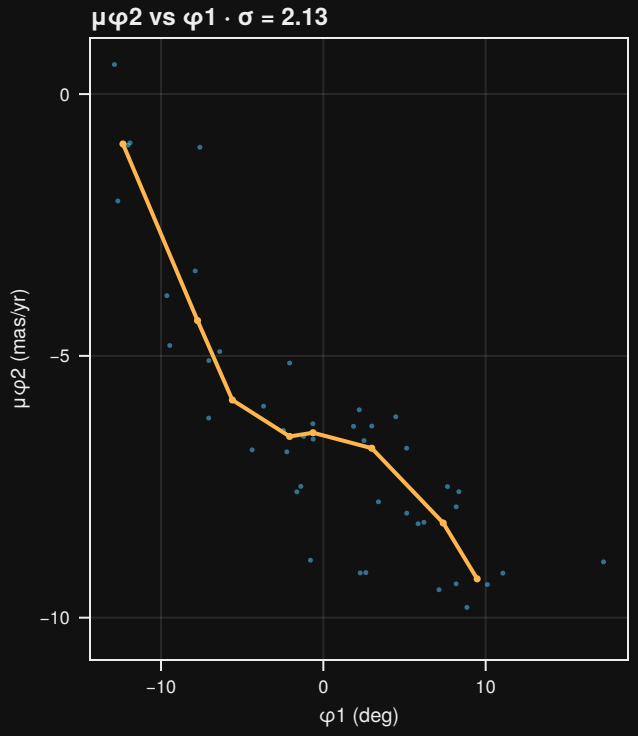


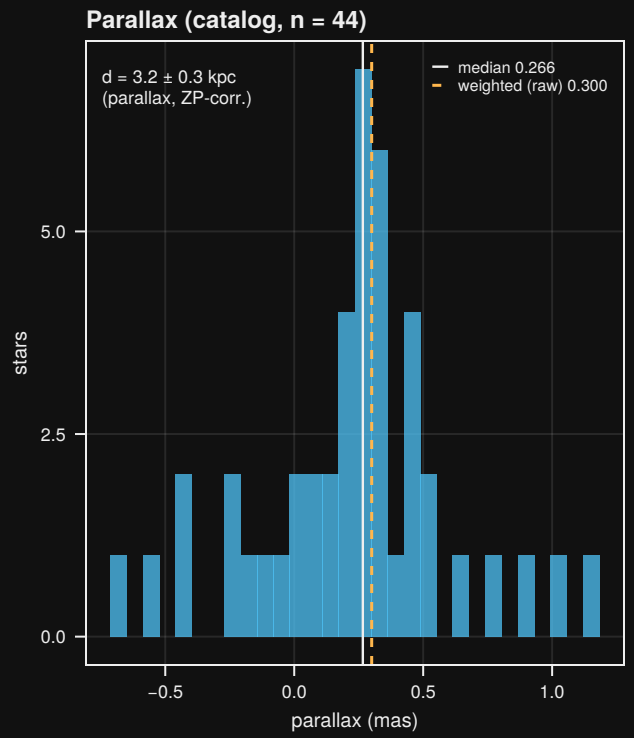
New-23

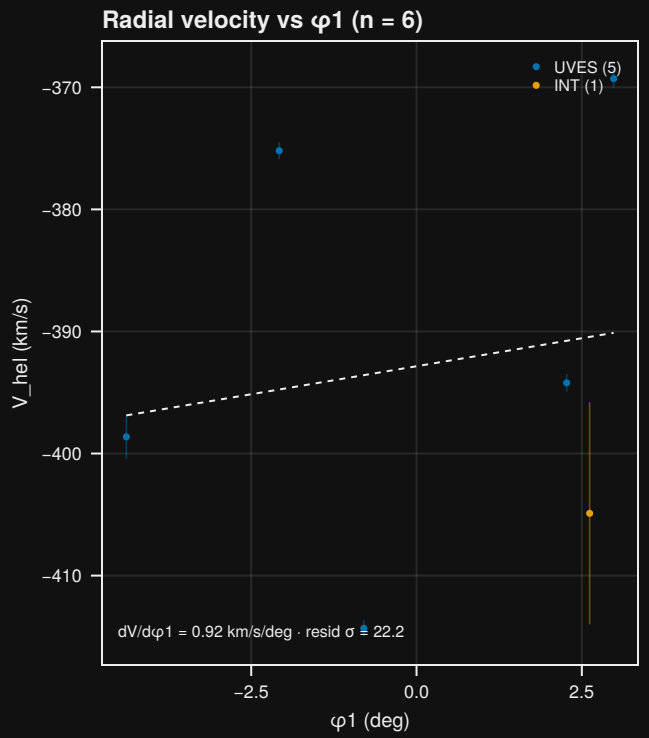
Kinematics · proper motion · parallax · radial velocity

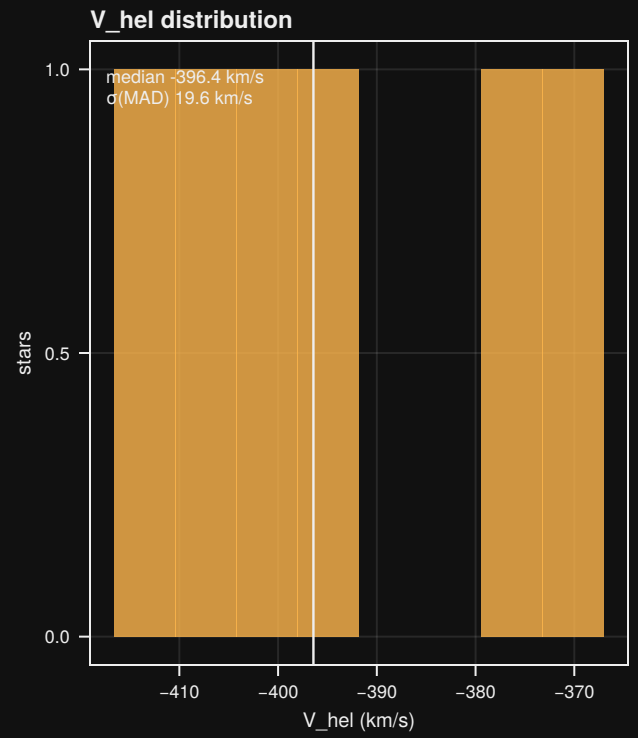






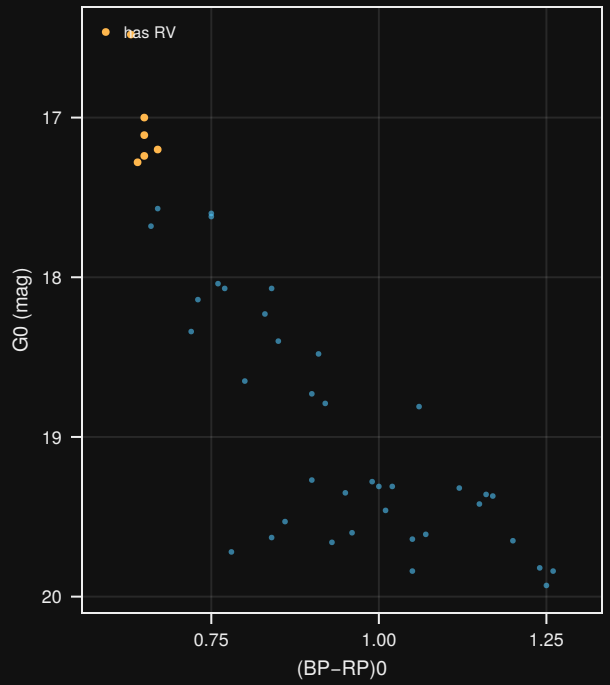




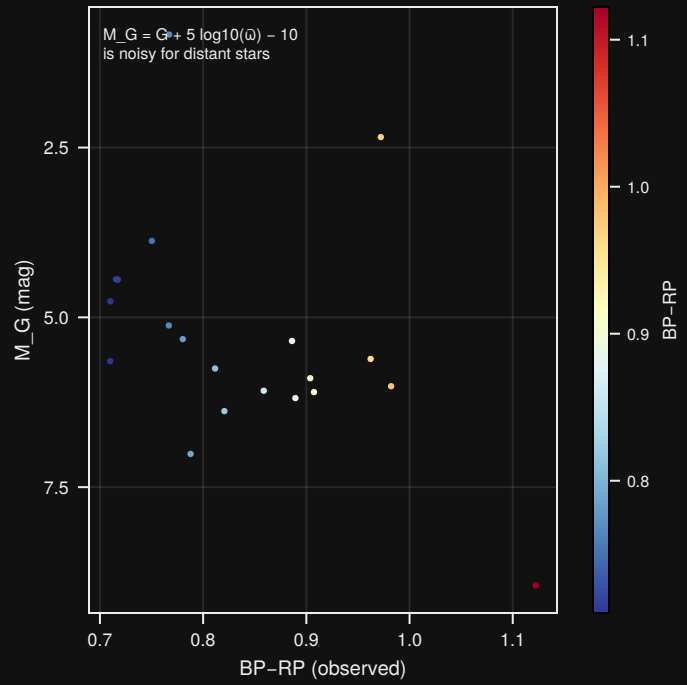


New-23

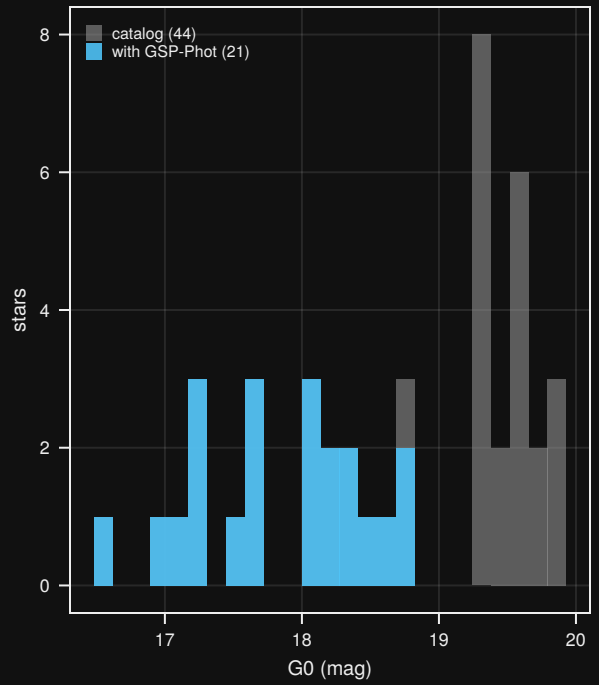
CMD · dereddened (n = 44)



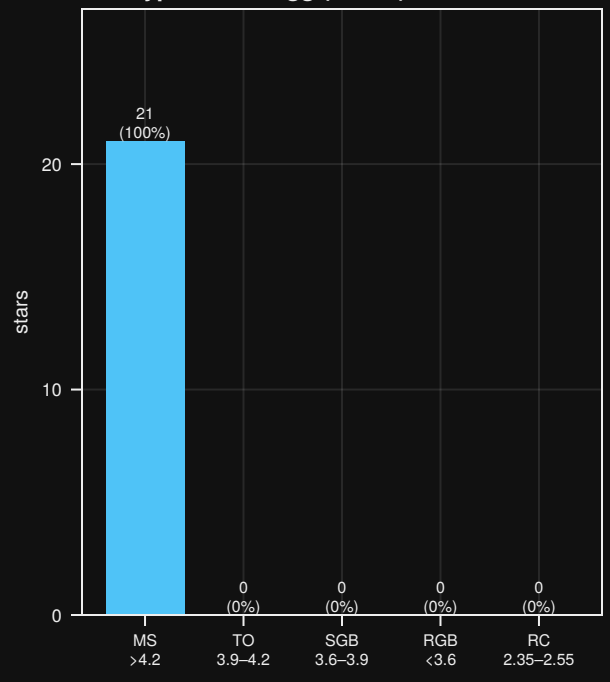
HR diagram · ϖ>0 (n = 20/21)



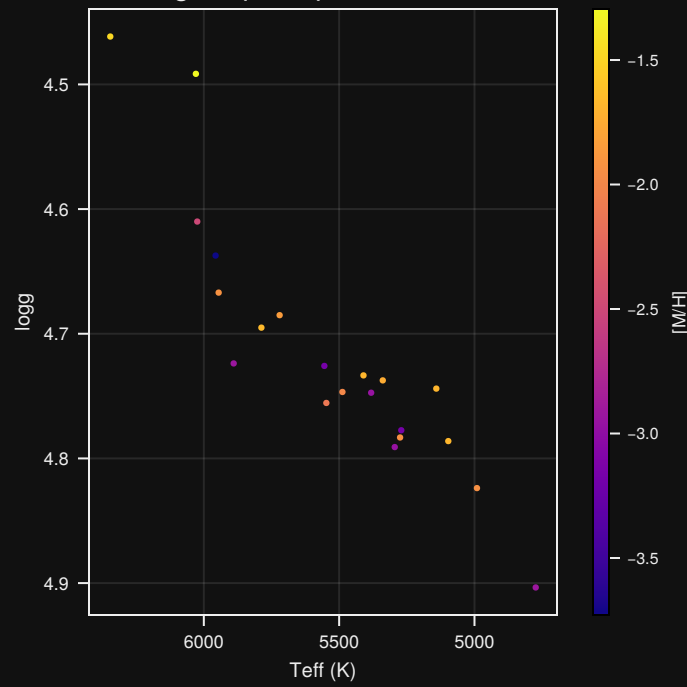
G0 & GSP-Phot coverage



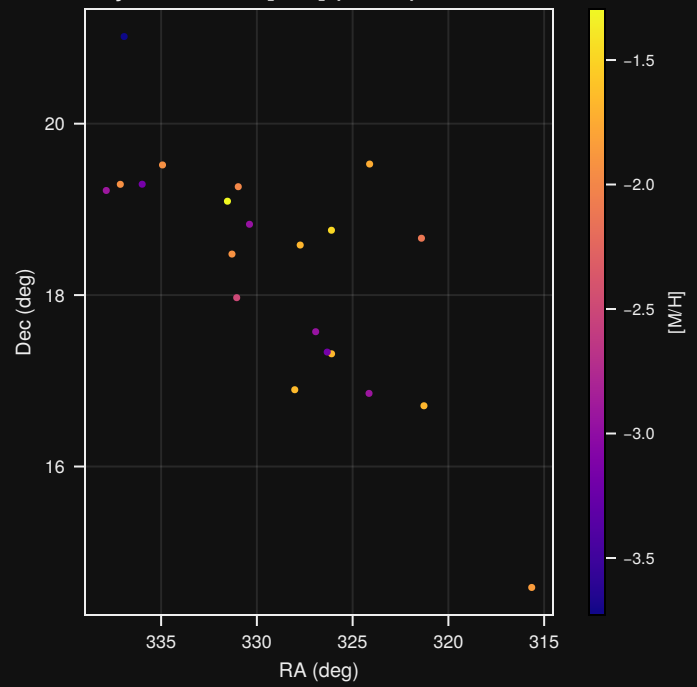
Star types from logg (n = 21)



Kiel diagram (n = 21)



Sky · GSP-Phot [M/H] (n = 21)

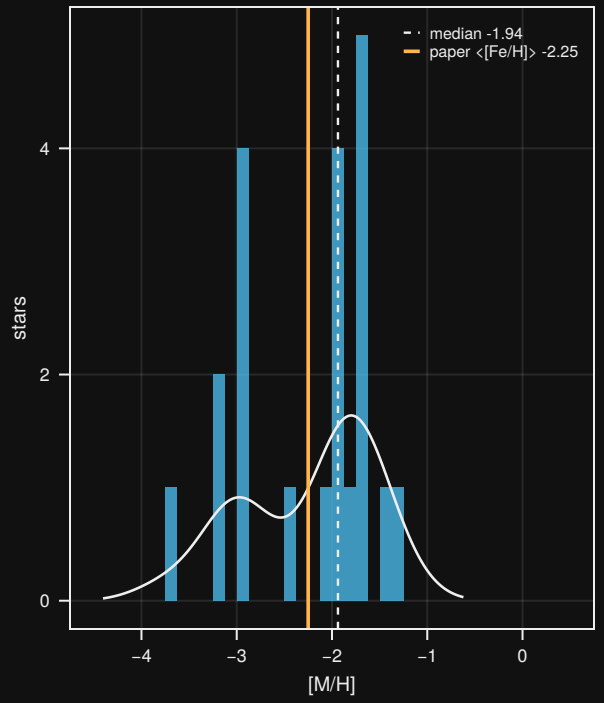


GSP-Phot (Gaia low-resolution BP/RP) parameters are prior-dominated for faint, distant halo stars (G ≥ 17): logg, [M/H] and ages are indicative only and typically biased toward dwarfs.

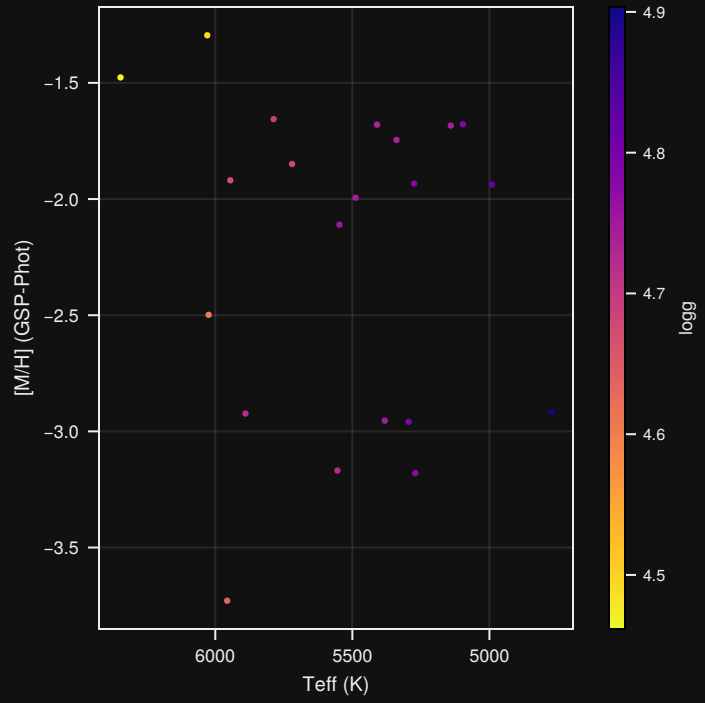
New-23

Metallicity · age · astrometric quality

[M/H] GSP-Phot (n = 21)



Teff vs [M/H] (n = 21)



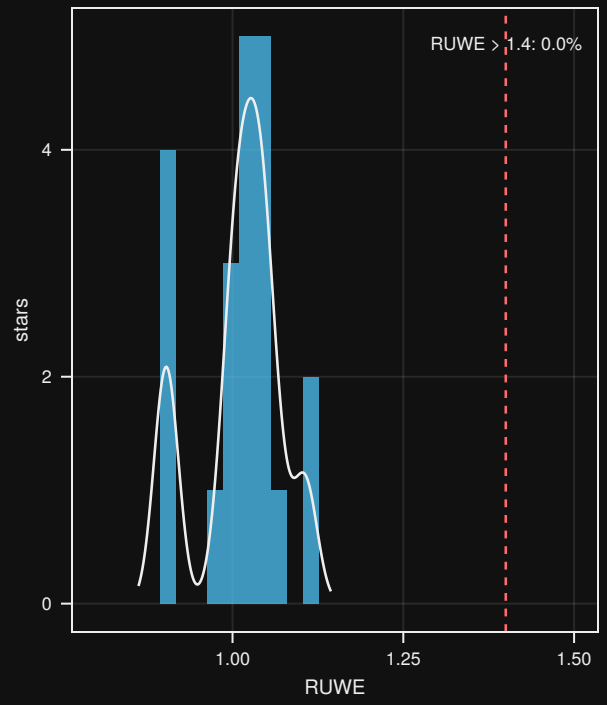
FLAME age (n = 0)

FLAME ages for only 0 star(s)
(FLAME needs good parallaxes)

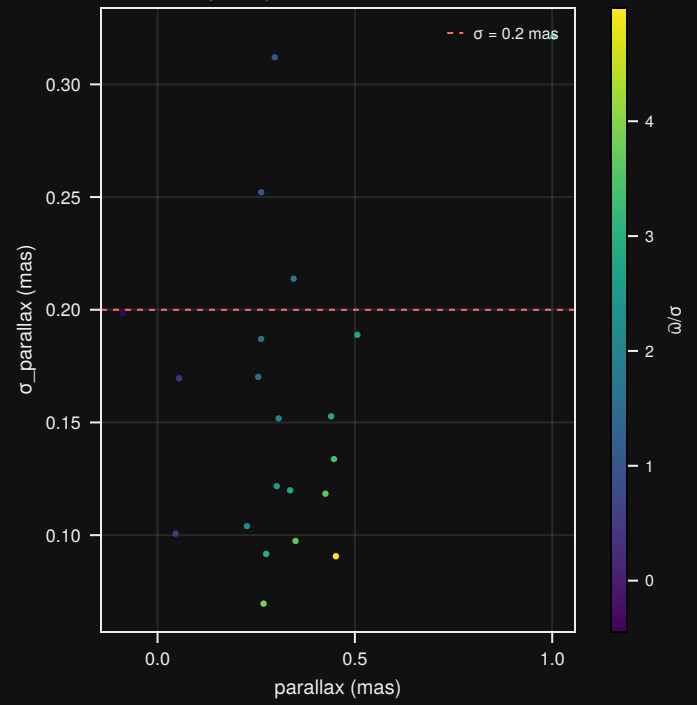
Age–metallicity (n = 0)

n < 3

RUWE (n = 21)

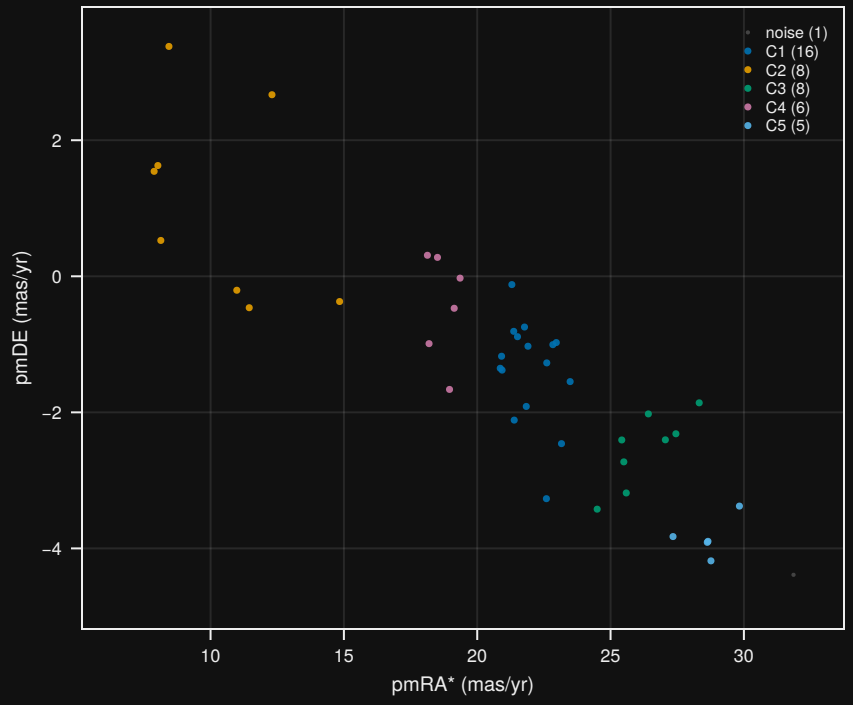


Parallax quality (n = 21)

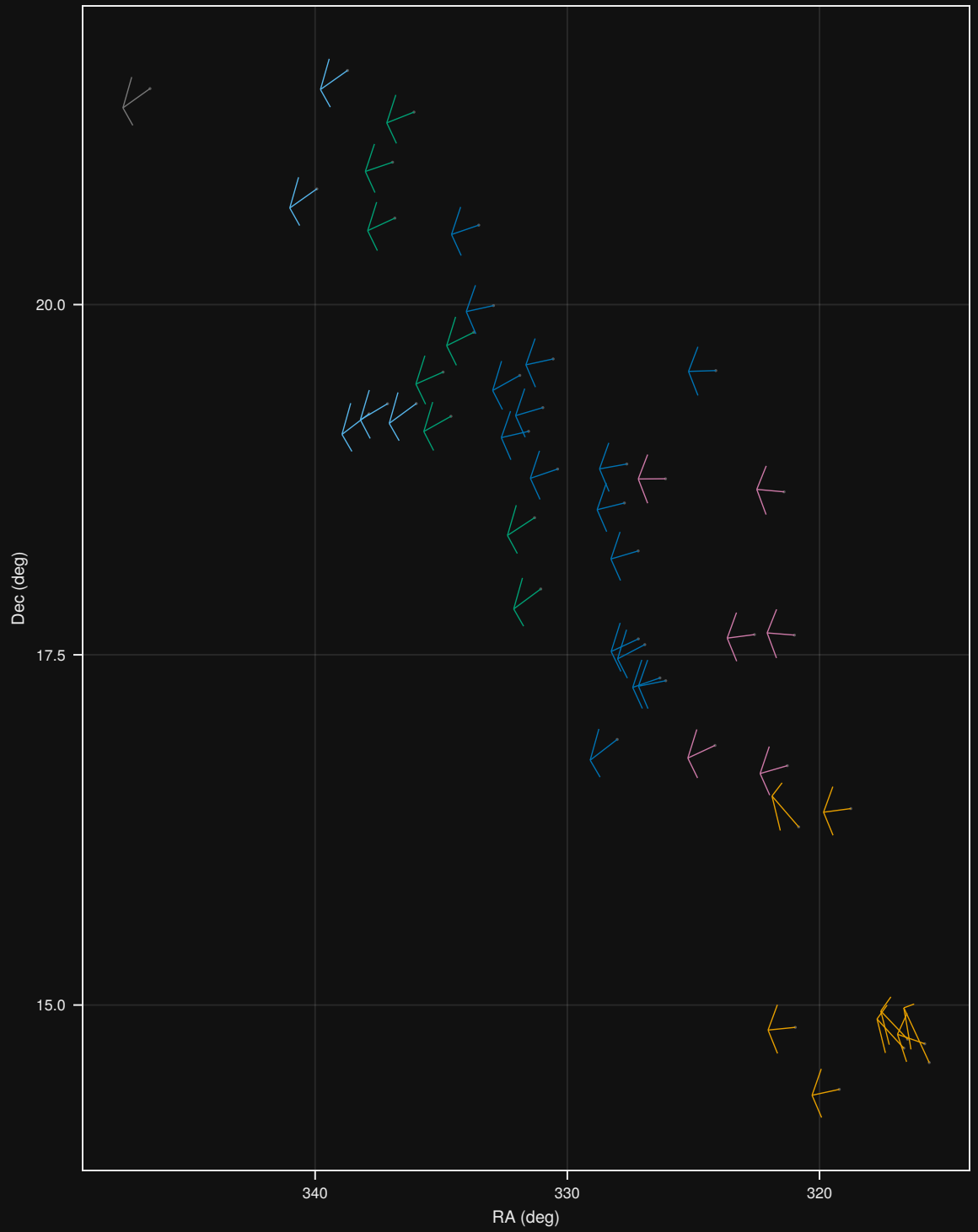


New-23

PM clustering · HDBSCAN · 5 cluster(s) · noise 2%



Proper-motion directions (arrows) coloured by cluster



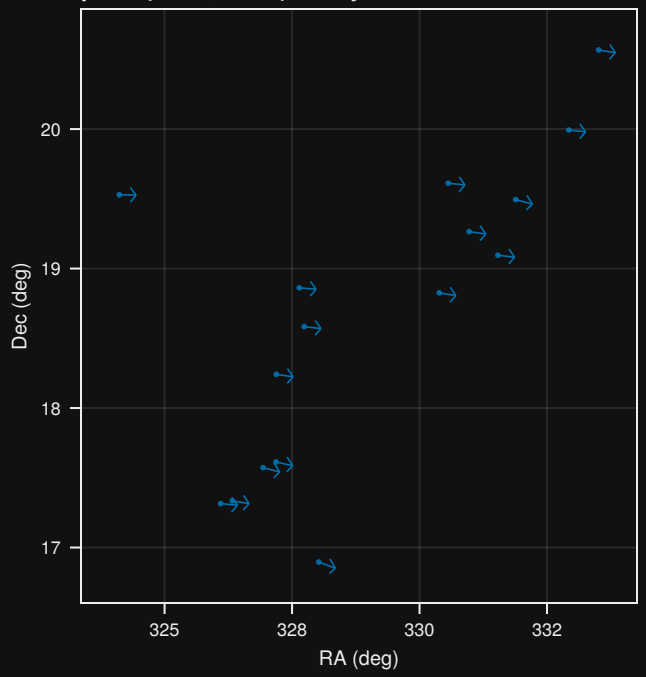
Cluster summary

id	n	pmRA	pmDE	σpmRA	[M/H]	n_RV
noise	1	31.86	-4.39	0.00	—	0
C1	16	21.81	-1.22	1.18	-1.75	2
C2	8	9.71	1.03	2.54	-1.85	0
C3	8	26.00	-2.41	1.21	-2.22	2
C4	6	18.73	-0.25	0.70	-1.90	2
C5	5	28.65	-3.90	0.17	-2.92	0

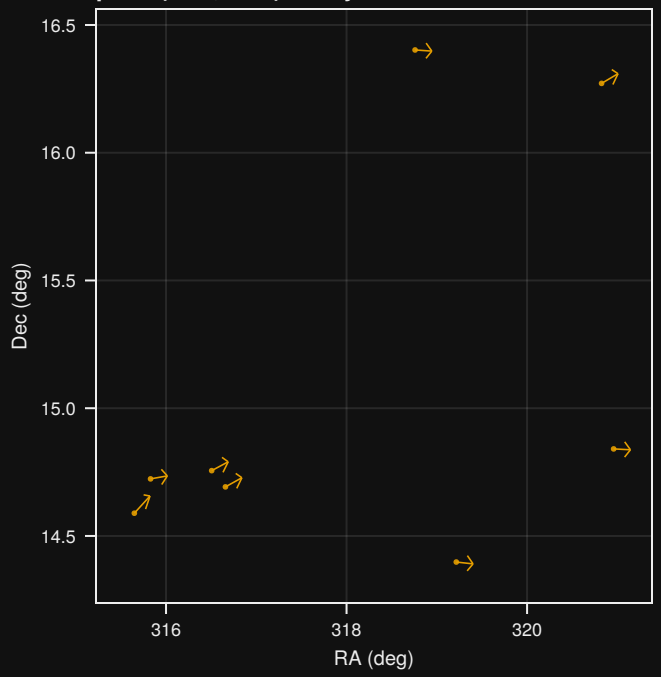
New-23

Per-cluster proper-motion vector maps

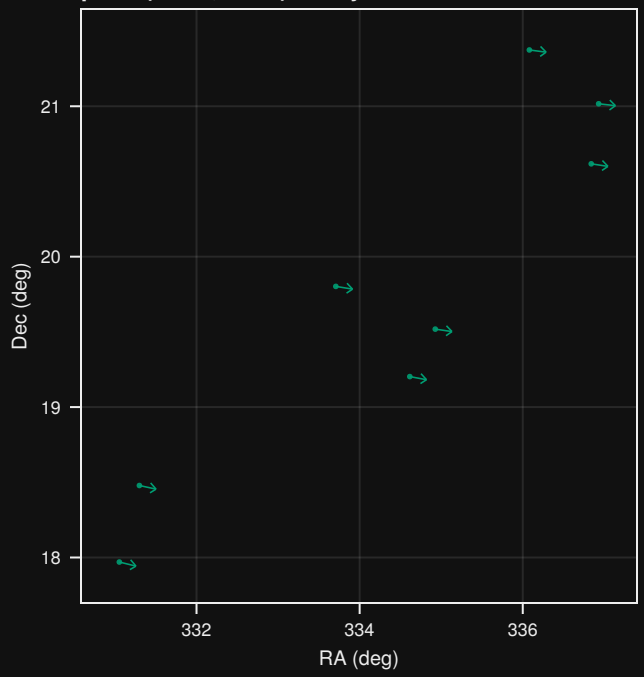
Cluster C1 · n = 16
pm = (21.81, -1.22) mas/yr



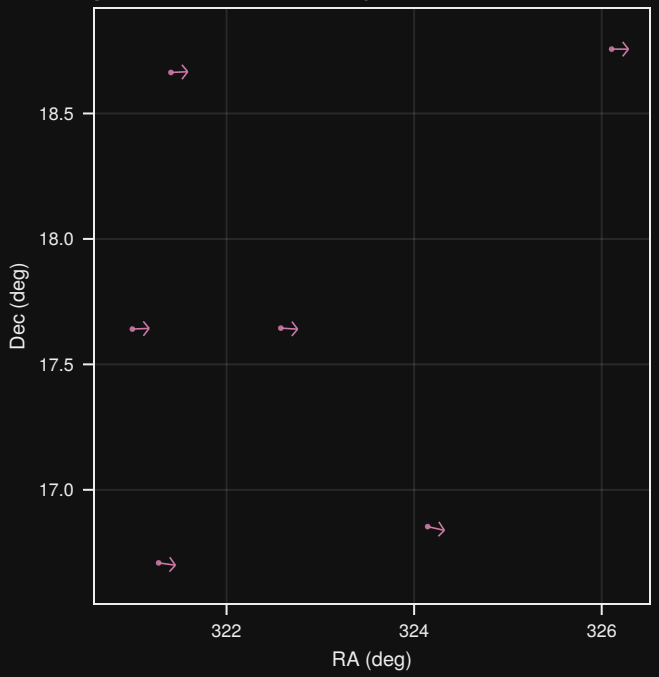
Cluster C2 · n = 8
pm = (9.71, 1.03) mas/yr



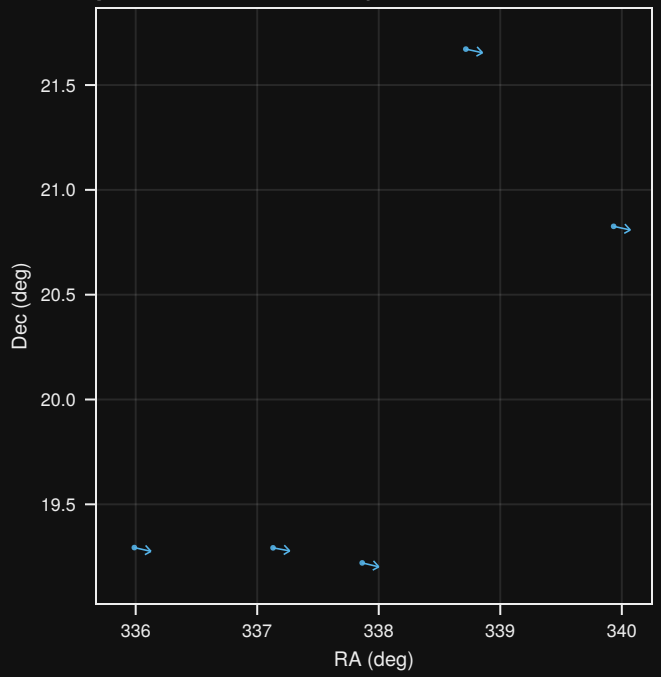
Cluster C3 · n = 8
pm = (26.00, -2.41) mas/yr



Cluster C4 · n = 6
pm = (18.73, -0.25) mas/yr



Cluster C5 · n = 5
pm = (28.65, -3.90) mas/yr



Noise · n = 1

